

Data TYPES

Def: Data type defines the kind of value a variable holds (no. text, decimal, true/false, etc.) & how much memory it uses.

Aspect	Details
How to use	Declare type before variable name → <code>int age = 20; double marks = 89.5</code>
when to use	pick based on the kind of value: whole no → int, decimal → double, character → char, true/false → boolean, text → string.
where to use	every variable declaration, method parameters, return types, arrays.
why specific type matters	Different types use different memory (int = 4 bytes, double = 8 bytes); prevents wasted memory & logical errors; no. type support math operations
Real-life example	Like different containers for dif ingredients - a jar for sugar, a bottle for oil. Each type "container" suits a specific kind of data.

Two categories

- ① primitive types - store actual values directly.
ex :- byte (1 byte), short (2 bytes), int (4 bytes), long (8 bytes), float (4), double (8), char (2), boolean (1 bit)
- ② non-primitive (reference) type - stores a reference to an object: String, arrays, classes, interfaces

Example program :-

```
int age = 20;
```

```
double marks = 89.5;
```

```
char grade = 'A';
```

```
boolean passed = true;
```

```
String name = "Rahul";
```

```
long population = 8000000000L //needs L suffix
```

```
float pi = 3.14f; //needs f suffix
```

Why 8 Primitive types (not one universal type):

Memory efficiency - picks the smallest type that fits the data.

Why long needs L: whole number literals default to int; L tells Java to treat it as long before overflow happens.

Why double is default over float: More precision (64-bit vs 32-bit); most calculations need that accuracy.

Date: / /



why string isn't primitive: text length varies, but primitives are fixed-size - so string is an object that manages a dynamic character array internally.

Real-life mapping:

Data type	Real-world data
int	Age
double	Height (5.8 ft)
char	Name initial
boolean	Light on/off
String	Full name
long	world population